ET AI HACKATHON: AGENTIC EDITION · SOLUTION DECK

Renewable Energy Orchestrator

An autonomous AI control room that reads satellite images, thermal inspections, documents and live data, simulates 100 futures for every decision, and runs an industry's solar, wind, batteries, loads and grid connection every 15 minutes — cheaper, cleaner and safer.

-23%	-42%	-56%	0	D3 × F3
energy cost	CO_2	critical load cut	safety violations	claimed position

vs business-as-usual control, 100 randomized simulated days with unannounced storms, breakdowns and price spikes

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The problem

Industrial renewable portfolios must be re-planned every 15 minutes under deep uncertainty

Volatile, messy inputs

- Solar & wind swing with clouds and storms
- IEX prices change every 15-min block
- Crucial facts arrive as images, PDFs and emails: inspection photos, work orders, grid notices

Conflicting goals

- Lowest cost vs cleanest energy vs never cutting critical load
- The right trade-off depends on the company's risk appetite

Today's practice

- Rule-of-thumb control: charge on surplus, discharge on deficit
- Reacts only to *now*; ignores forecasts, prices, documents and risk
- Breaks grid-line limits, wastes clean power

Business-as-usual on our test portfolio (5 solar, 3 wind, 2 batteries, 4 industrial loads)

- ₹94.2 lakh per day energy cost · 957 t CO_2 per day · only 62% clean energy
- 12.1 MWh of **critical** load cut per day on average (on 20 of 100 days) · 4.9 physical-limit violations per day

Our solution at a glance

An agent that reads heterogeneous inputs, reasons over all risks at once, simulates futures, acts autonomously and learns

Perceive
live data + images + documents

Diagnose
9 risk signals at once

Plan
33 plans from 9 recipe bands

Simulate
each plan × 100 futures

Choose
4-rule ladder ± LLM critique

Verify & act
physics validator

Explain
reason + audit log

Learn
forecast bias & bands

What makes it different

- Reads **satellite images, thermal inspection images and PDF / text notices** and turns them into validated, confidence-scored facts
- No fixed playbook: every risk counts simultaneously, as a probability
- The LLM can propose and test its own plan, but never types a megawatt — maths computes, physics validates
- Autonomous inside management-set limits, with operator override and review

Built and working

- Streamlit control room: industry setup → live operations → results & learning
- Live Open-Meteo weather + NASA satellite scenes for the real farm coordinates
- MILP optimiser, Monte Carlo stress test, online learning, multimodal intake
- Labelled reliability tests, 90+ automated tests, 100-day Monte Carlo

Our position in the 3×3 grid: D3 × F3

Self-assessment

Highest depth and highest feature level — each axis evidenced below

	F1	F2	F3
D1	covered	covered	covered
D2	covered	covered	covered
D3	covered	covered	**CLAIMED**

F1 optimal action cluster · F2 + situation-specific criteria under uncertainty · F3 + simulation over time with varying inputs
D1 acceptable outputs · D2 structured input, high reliability · D3 heterogeneous multimodal input, high reliability

Axis	Requirement	Evidence
D3	Highly heterogeneous multimodal input	Satellite images (NASA MODIS) · thermal / drone images · PDF work orders & grid notices · free-text fault reports & emails · weather API (JSON) · price & consumption CSV · config tables
D3	High degree of demonstrable reliability	0 wrong of 256 applied findings, from 350 labelled images and documents · 100% of vague notices quarantined · 0 safety violations in 100 days · cheaper than business-as-usual on 100/100 days
F3	Simulate over time with varying inputs	100 sampled futures every 15 min · 24 h day-ahead plan · full-day digital twin with 11 shock types · 100-day Monte Carlo

What we deliberately do not claim

- Audio / voice input is not supported yet
- Thermal inspection images are synthetic with known ground truth (no real drone data was available); satellite scenes, weather data and document formats are real
- Real drone images without temperature-calibration metadata are analysed but held for operator review until calibrated with the camera's temperature range (by design)
- Plant meters come from a physics-based digital twin; the same interface accepts live SCADA

D3: highly heterogeneous multimodal input

D3

Every modality becomes a confidence-scored finding before it can touch a decision

Modality	Input	How it is understood	What it changes
Satellite scenes	NASA GIBS MODIS true-colour image around each solar farm	Cloud mask (white & unsaturated pixels) → measured cloud % vs forecast at overpass	≥25-pt gap → solar –/+ around overpass
Thermal / drone images	Radiometric PNG/JPG of PV blocks	Per-module median → noise filter → robust threshold → hot-spot regions → modules	Farm derate + repair job the AI schedules
PDF & text notices	Work orders, fault reports, SLDC grid notices, production notices, emails	Classify → extract asset, time window, size (rules, or LLM when configured) → schema validation	Maintenance request, outage, line limit or demand change
Structured feeds	Weather API (JSON), IEX prices, consumption CSV, asset tables	Live → last-year proxy → saved day → simulated fallback	Forecasts & profiles

Read
image / PDF / text / API

Extract
CV, parser or LLM

Validate
schema, ranges, asset ids, image quality

Gate
≥75% confidence → apply; else quarantine

Act
twin updated, AI re-plans instantly

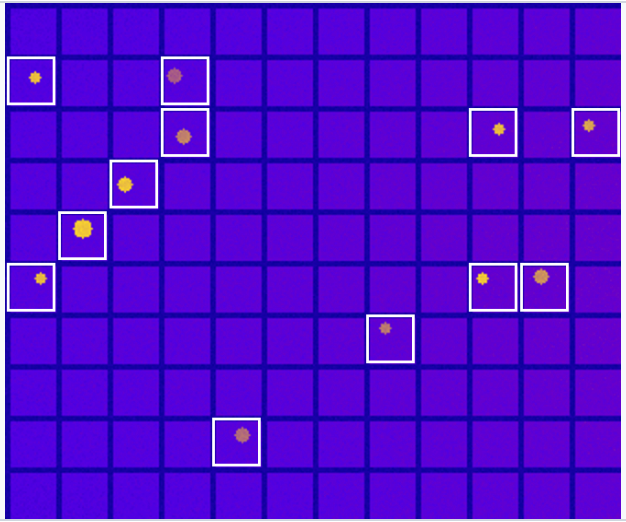
D3 in action: what the agent actually reads

D3

A real satellite scene, a thermal inspection and a grid notice — each turned into a decision-ready finding



🛰️ Bhadla (RJ) · NASA MODIS Terra · 2026-08-23 · detected cloud 37%. Left: raw NASA scene; right: detected cloud edge (orange). Compared with the forecast at the 10:30 overpass; a ≥25-point gap changes the solar plan.



🌡️ Drone thermal image of a PV block (synthetic, known truth). White boxes = modules the agent flagged → **S2: 12 of 120 modules with hot-spots (10.0%) → capacity –5.0%; inspection crew needed**

```
STATE LOAD DESPATCH CENTRE
GRID OPERATOR NOTICE
Ref: SLDC/OPS/2026/1187

Due to transmission congestion on the 220
kV evacuation corridor, the
interconnection import limit for the
consumer will be reduced to 60% from
17:30 for 3 hours.
```

📄 PDF notice → extracted **line limit 60%, 17:30–20:30, confidence 100% → applied**. A vague email ("may possibly take W1 or W3 down… to be confirmed") → **quarantined**.

D3: demonstrable reliability on labelled data

D3 · reliability

Measured accuracy, and — most important — nothing wrong is ever acted on

Thermal tier	Images	Exact defect count	Applied	Wrong applied
Easy	50	100%	49	**0**
Medium	50	100%	47	**0**
Hard	50	16%	0	**0**

Hard tier = heavy sensor noise + sun glare: detection is unreliable, so every one is quarantined instead of applied.

Satellite cloud detector	Result
Test	60 synthetic clouds over 9 real NASA scenes of our sites
Mean error	0.33 % points (max 2.5)

The detector is a whiteness threshold, so this controlled test is favourable to it; real hazy or thin cloud is harder, which is why the satellite reading is cross-checked against the forecast before acting.

Documents (200 notices)	Result
Clear notices applied correctly	160 / 160 (100%)
Vague / tentative notices quarantined	40 / 40 (100%)
Wrong findings applied	**0**
Formats covered	24 h, 12 h am/pm, 'hrs', durations, ranges, dates, filler text, PDF

Safety metric across all modalities

0 wrong of 256 applied findings, from 350 labelled images and documents. Corrupt or unsupported files are rejected without interrupting control.

Feature evidence: F1 → F2 → F3

F3

Each level is demonstrated live in the prototype and verified by tests

Level	Requirement	How the agent does it	Where to see it
F1	Determine optimal cluster of actions	MILP jointly sets battery charge / discharge, grid import / export, curtailment per farm and demand response per load — one consistent action set every 15 min	Live tab → action log
F2	Optimal options after determining optimality criteria for the situation, adjusting for uncertainty	9 continuous signals + multimodal evidence → 33 plans from 9 recipe bands (weights, reserve, forecast caution) → ranked against the industry's risk tolerance	'What the AI is thinking' panel
F3	Simulate the scenario over time, accounting for variations in input conditions	Every plan replayed in 100 futures (forecast spread + storm probabilities) over 4 h; 24 h day-ahead plan; full-day twin with 11 shocks; 100-day Monte Carlo	Inject a storm at 10% vs 90%

Graded response to uncertainty (same day, 18:00 storm forecast): 0% storm → 27% battery reserve · 10% storm → 43% battery reserve · 50% storm → 52% battery reserve · 90% storm → 60% battery reserve

Architecture

Process flow from heterogeneous inputs to autonomous action — every block is implemented

Industry inputs
loads, assets, limits, maintenance, targets, risk appetite

Images
NASA satellite scenes · thermal / drone inspections

Documents
PDF work orders · SLDC notices · fault reports · emails

Live data
Open-Meteo 15-min weather · IEX prices · plant state (twin / SCADA)

Multimodal intake
extract → validate → confidence gate (≥75%) → twin

→

Autonomous agent loop · every 15 min + instantly on any shock

1 Perceive	state, learned P10/P50/P90 forecasts, alerts with probabilities, evidence
2 Diagnose	9 risk signals simultaneously
3 Options	9 recipe bands → explore 27 + refine 6 = 33 plans (MILP, PuLP/CBC)
4 Stress-test	each plan × 100 futures → avg ₹, worst ₹, P(shortfall), CO_2
5 Choose	ladder: safe → risk ≤ tolerance → cost/clean → battery wear; LLM may propose & pick
6 Verify	physics validator; repair ×2 → safe fallback
7 Act + explain	execute, JSONL audit log, plain-English reason
8 Learn	forecast bias & band width, saved daily

→

LLM (optional, any provider)
Claude · Groq · Gemini · OpenRouter · Ollama. Critiques the pick, proposes its own plan (tested like all others), explains. Never outputs MW.

Day-ahead backbone
24 h LP battery targets, re-solved after shocks

Human oversight
operator override → safe mode · high-impact review · quarantine queue

Outputs
set-points · ₹ & CO_2 saved · clean % · efficiency · audit log

How the AI chooses: 33 plans × 100 futures × 4 rules

Optimisation, simulation and a transparent priority ladder — under a second per decision

Recipe band	Battery reserve	Forecast caution	Price / clean / reliability
Max savings	5–15%	0–15%	≈65 / 20 / 15
Lean	12–25%	0–25%	≈59 / 19 / 22
Balanced	22–35%	10–35%	≈53 / 18 / 30
Green-lean	20–40%	0–25%	≈30 / 55 / 15
Steady	32–45%	20–50%	≈46 / 16 / 38
Cautious	42–55%	35–65%	≈39 / 15 / 46
Protective	52–68%	50–80%	≈32 / 13 / 55
Strong reserve	64–80%	65–95%	≈24 / 12 / 65
Fortress	76–90%	80–100%	≈16 / 10 / 73

3 combinations explored inside every band (27) + 6 refined around the best = 33 MILP plans, replayed in the same 100 futures (a 35% storm hits in ~35 of them).

1 · Safety

Must pass the physics validator (state of charge, rates, line limits, balance, availability).

2 · Reliability

Chance of cutting critical load ≤ the industry's tolerance (default 2%).

3 · Cost vs clean

Lowest score = avg ₹ + risk-aversion × (worst – avg) + carbon-pressure × CO_2 value.

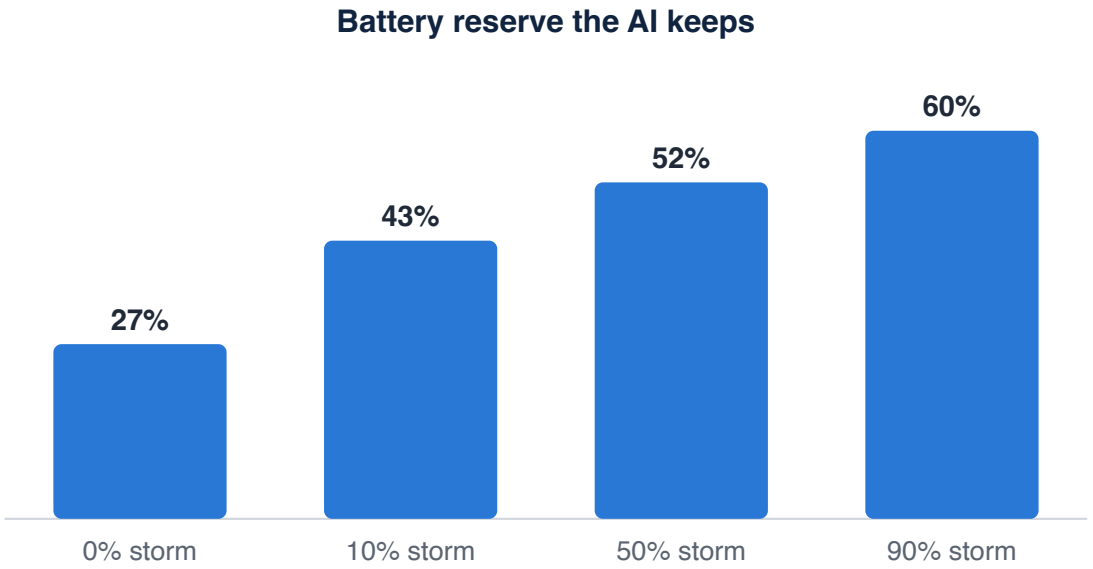
4 · Battery life

Among near-equal plans, least battery cycling wins.

Adjusting for uncertainty: protection scales with probability

F2 · F3

Same day, same time — only the storm probability changes



Why it matters

A fixed playbook treats every storm alert alike. Here the storm is sampled into ~p% of 100 futures, so protection rises gradually with risk — the AI pays for protection only when the expected outage cost justifies it.

What the agent says (real output)

I tested 33 combinations from 9 recipe bands across 100 possible futures and chose a 'Protective' plan (keep ≥52% battery reserve) … 'Max savings' would save money but risks a critical shortfall in 50% of futures (limit 2%).

Combined risks are handled together: a storm, a price spike, a battery fault and a grid notice all enter the same 100 futures — nothing is collapsed into one 'situation'.

Agentic & autonomous capability

Agentic

Reasons, plans, uses tools, acts, recovers from errors and learns

Reasons

Diagnoses 9 interacting risk signals plus image & document evidence; writes its own rationale.

Plans

24 h day-ahead plan + rolling 4 h plans, re-planned every 15 min and instantly after shocks or new documents.

Uses tools

MILP optimiser, Monte Carlo simulator, physics validator, image analysers, document parser, weather / satellite / price APIs, crew dispatch.

Acts autonomously

Dispatches assets, reduces flexible load, reschedules maintenance, sends repair crews after breakdowns or thermal hot-spots.

Recovers from errors

Infeasible → relax ×2 → safe fallback · API down → last-year proxy → replay → simulated · LLM failure → optimiser decides · bad files → rejected.

Learns & self-critiques

Learns forecast bias daily; the LLM can challenge the ladder by proposing its own plan, which must win on the same stress test.

Effective use of AI: the right technique for each job

AI

AI is the core of every decision, not a bolt-on

AI technique	Role in the agent	Why this technique
Computer vision	Hot-spot detection in thermal images; cloud masking in satellite scenes	Turns pixels into capacity and weather facts, with quality checks
Document understanding (parser / LLM)	Work orders, grid notices, fault reports → structured events	Operational reality arrives as text; validated before use
Mathematical optimisation (MILP)	Each plan → exact feasible MW for 4 h	Physically consistent numbers — no hallucination possible
Probabilistic simulation	100 futures per decision from forecast quantiles & alert probabilities	Honest risk: average, worst case, probability of outage
Search	Explore + refine across 9 recipe bands	Finds in-between optima a fixed menu misses
Online machine learning	Forecast bias & calibration from every outcome	Learned wind bias -9.1% (true -10.7%)
LLM agent (any provider)	Critiques the recommendation, proposes & tests its own plan, explains	Reasoning & communication, fenced by code

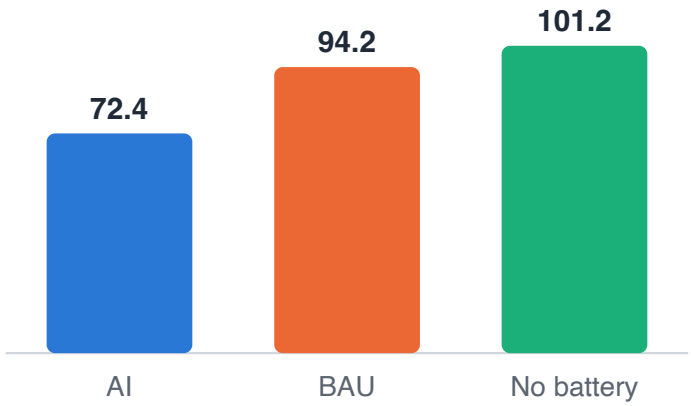
Transparency: the results in this deck were produced with the optimiser + ladder (offline mode). The LLM layer is optional, provider-neutral (incl. free Groq / Gemini tiers) and the system never depends on it.

Business impact: 100-day Monte Carlo

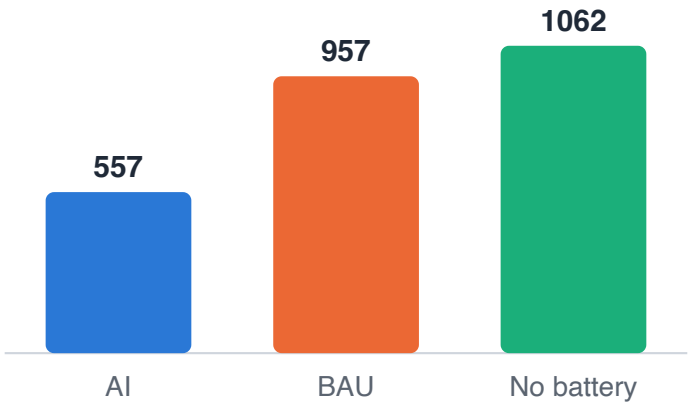
Impact

Randomized days with 3 unannounced shocks each; all controllers see identical weather, prices and events

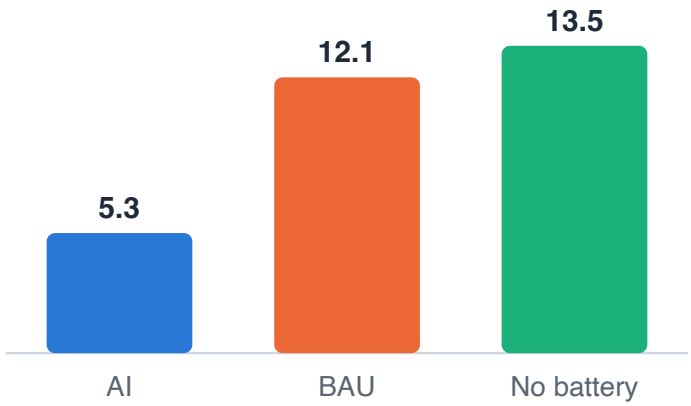
Cost (₹ lakh / day)



CO_2 (t / day)



Critical load cut (MWh / day)



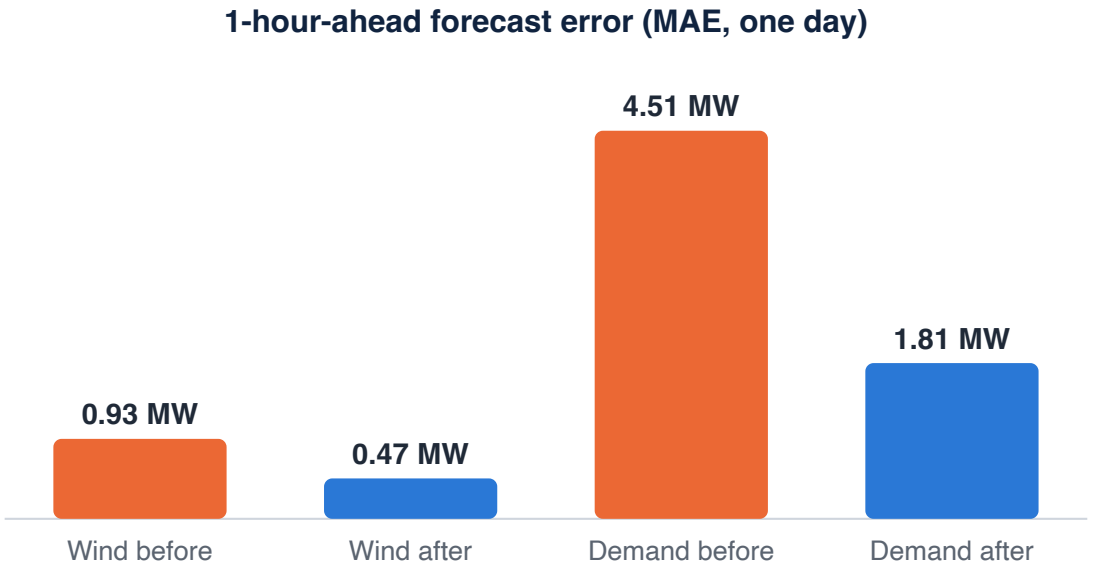
Mean per day	AI agent	Business-as-usual	Improvement
Energy cost	₹72.4 L	₹94.2 L	**-23%** · cheaper on 100/100 days
CO_2	557 t	957 t	**-42%**
Critical load cut	5.3 MWh	12.1 MWh	**-56%** · days affected 8 vs 20
All load cut (incl. flexible)	5.4 MWh	30.1 MWh	-82%
Clean energy share / safety violations	72.4% / 0.00	61.6% / 4.87	+10.7 pts / zero

We lead with critical load (-56%) rather than total load (-82%): nearly all of the AI's remaining cuts are physical shortages (e.g. storm + congested line), where it protects critical load first and says so.

Learning: the agent gets better every day

Trainability

Online correction of forecast bias and uncertainty bands



Forecast	Hidden model bias	Bias the agent learned
Wind	-10.7%	-9.1%
Demand	-2.9%	-3.3%

- Every tick, the 1-hour-ahead forecast is compared with what actually happened
- Bias learned with a robust moving average; P10–P90 bands widened or narrowed to ~80% coverage
- Lessons saved at end of day and used from the first tick tomorrow
- Honest note: solar error did not improve on the test day (sunrise / sunset ramps dominate)

Working prototype

Streamlit control room — runs without internet or API key; live data when available

Demo

Evidence from images & documents



Best recipe & why

1 · Your industry

Loads & consumption CSV, solar / wind / batteries with coordinates, grid limits, maintenance & breakdowns, targets, satellite / image / document inputs

2 · Live control room

Status, ₹ & CO_2 saved, all risk signals, every plan considered, evidence table, high-impact review, inject 11 events or feed a document

3 · Results & learning

Best recipe & why, AI vs business-as-usual, learning table, Monte Carlo proof

Responsible AI, human oversight & robustness

Responsible AI

Autonomous by design — governed by limits, verification, override and audit

No hallucinated actions

The LLM can only propose settings or pick a validated plan id; all MW come from the optimiser; ineligible picks are rejected in code.

Multimodal guardrails

Every image / document finding is validated and confidence-scored; below 75% it is quarantined for a human, never applied.

Human oversight

Operator override hands control to the safe controller instantly; high-impact actions are flagged for review; management sets risk limits and recipe ranges.

Physics guardrails

Validator checks state of charge, rates, line limits, balance, critical load and availability before every action; the twin independently counts violations.

Transparency & audit

Every decision logged: signals, evidence, 33 options, stress results, ladder, action, explanation — reproducible with seeded scenarios.

Security & privacy

No personal data; keys only in .env; read-only SCADA first; OT isolation; shadow-mode rollout before control.

Scalability & path to a real plant

Scale

Industry-agnostic by configuration; one interface swaps the twin for live data

Week 0 · Configure
Assets, loads, schedules, targets (UI, YAML or CSV); upload documents & images.

Weeks 1–2 · Shadow mode
SCADA / meters / BMS read-only (OPC-UA, Modbus, MQTT); AI recommends, operators compare.

Weeks 3–4 · Battery control
Battery set-points only; watchdog returns to local control on silence.

Month 2+ · Full autonomy
Curtailment & demand response; learning tunes forecasts per site.

- Any number of farms, batteries and loads — sites without wind or batteries are tested
- < 1 s per decision for 33 plans × 100 futures on a laptop; solves run in parallel
- Same agent for steel, cement, textiles or data centres — only configuration changes
- New modalities plug into the same extract → validate → gate → apply pipeline

Honest limitations & roadmap

What we have not proven yet, and what comes next

- Thermal images are **synthetic** with known ground truth (no real drone data available); satellite scenes are real
- No audio / voice input yet
- Plant meters, batteries and demand come from a physics-based digital twin; weather and satellite are live
- IEX price file is a labelled sample
- Results produced in offline mode; the LLM's added value is not yet measured on the 100 days
- Band search finds very good plans, not a proven global optimum

- Real drone / thermal datasets and panel-defect models
- Voice notes from field crews → the same document pipeline
- Live SCADA adapter with data-health monitoring
- Real IEX DAM + RTM prices and DSM penalty rules
- A/B the LLM layer over 100 days with a free-tier model
- Full stochastic optimisation as the end-state optimum

Cheaper, cleaner, never dark — decided by an AI that shows its work.

Claimed position **D3 × F3** · 100-day proof: -23% cost, -42% CO_2, -56% critical load cut, 0 safety violations · 0 wrong multimodal findings applied
Code & README: github.com/Akshadman/renewable-energy-orchestrator

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